LECTURE 3
GEOMETRY OF LS, PROPERTIES OF $\widehat{\sigma}^2$, PARTITIONED REGRESSION, GOODNESS OF FIT

Geometry of LS

We can think of y and the columns of X as members of the n-dimensional Euclidean space $\mathbb{R}^n$. One can define a subspace of $\mathbb{R}^n$ called the *column space* of an $n \times k$ matrix X, that is, the collection of all vectors in $\mathbb{R}^n$ that can be written as linear combinations of the columns of X:

$$\mathcal{S}(X) = \left\{ z \in \mathbb{R}^n : z = Xb, b = (b_1, b_2, \ldots, b_k)^\top \in \mathbb{R}^k \right\}.$$

For two vectors a, b in $\mathbb{R}^n$, the distance between a and b is given by the Euclidean norm[1] of their difference $\|a - b\| = \sqrt{(a-b)^\top (a-b)}$. Thus, the LS problem, minimization of the sum of squared errors $(y - Xb)^\top (y - Xb)$, is to find, out of all elements of $\mathcal{S}(X)$, the one closest to y:

$$\min_{\widetilde{y} \in \mathcal{S}(X)} \|y - \widetilde{y}\|^2 .$$

The closest point is found by "dropping a perpendicular". That is, a solution to the LS problem, $\widehat{y} = X\widehat{\beta}$, must be chosen so that the residual vector $\widehat{u} = y - \widehat{y}$ is orthogonal (perpendicular) to each column of X:

$$\widehat{u}^\top X = 0.$$

As a result, $\widehat{u}$ is orthogonal to every element of $\mathcal{S}(X)$. Indeed, if $z \in \mathcal{S}(X)$, then there exists $b \in \mathbb{R}^k$ such that $z = Xb$, and

$$\begin{aligned} \widehat{u}^\top z &= \widehat{u}^\top X b \\ &= 0. \end{aligned}$$

The collection of the elements of $\mathbb{R}^n$ orthogonal to $\mathcal{S}(X)$ is called the *orthogonal complement* of $\mathcal{S}(X)$:

$$\mathcal{S}^\perp(X) = \left\{ z \in \mathbb{R}^n : z^\top X = 0 \right\}.$$

Every element of $\mathcal{S}^\perp(X)$ is orthogonal to every element in $\mathcal{S}(X)$.

As we have seen in Lecture 2, the solution to the LS problem is given by

$$\begin{aligned} \widehat{y} &= X\widehat{\beta} \\ &= X \left(X^\top X \right)^{-1} X^\top y \\ &= P_X y, \end{aligned}$$

where

$$P_X = X \left(X^\top X \right)^{-1} X^\top$$

is called the *orthogonal projection matrix*. For any vector $y \in \mathbb{R}^n$,

$$P_X y \in \mathcal{S}(X).$$

Furthermore, the residual vector will be in $\mathcal{S}^\perp(X)$:

$$y - P_X y \in \mathcal{S}^\perp(X). \tag{1}$$

[1]For a vector $x = (x_1, x_2, \ldots, x_n)^\top$, its Euclidean norm is defined as $\|x\| = \sqrt{x^\top x} = \sqrt{\sum_{i=1}^n x_i^2}$.

To show (1), first note that since the columns of X are in $\mathcal{S}(X)$,

$$\begin{aligned} P_X X &= X \left(X^\top X\right)^{-1} X^\top X \\ &= X, \end{aligned}$$

and, since P_X is a symmetric matrix,

$$X^\top P_X = X^\top .$$

Now,

$$\begin{aligned} X^\top \left(y - P_X y\right) &= X^\top y - X^\top P_X y \\ &= X^\top y - X^\top y \\ &= 0. \end{aligned}$$

Thus, by definition, the residuals $y - P_X y \in \mathcal{S}^\perp(X)$. The residuals can be written as

$$\begin{aligned} \widehat{u} &= y - P_X y \\ &= \left(I_n - P_X\right) y \\ &= M_X y, \end{aligned}$$

where

$$\begin{aligned} M_X &= I_n - P_X \\ &= I_n - X \left(X^\top X\right)^{-1} X^\top \end{aligned}$$

is a projection matrix onto $\mathcal{S}^\perp(X)$.

The projection matrices P_X and M_X have the following properties:

- $P_X + M_X = I$. This implies that for any $y \in \mathbb{R}^n$,

$$y = P_X y + M_X y.$$

- Symmetric:

$$\begin{aligned} P_X^\top &= P_X, \\ M_X^\top &= M_X. \end{aligned}$$

- Idempotent: $P_X P_X = P_X$, and $M_X M_X = M_X$.

$$\begin{aligned} P_X P_X &= X \left(X^\top X\right)^{-1} X^\top X \left(X^\top X\right)^{-1} X^\top \\ &= X \left(X^\top X\right)^{-1} X^\top \\ &= P_X \\ M_X M_X &= \left(I_n - P_X\right)\left(I_n - P_X\right) \\ &= I_n - 2P_X + P_X P_X \\ &= I_n - P_X \\ &= M_X. \end{aligned}$$

- Orthogonal:

$$\begin{aligned} P_X M_X &= P_X \left(I_n - P_X\right) \\ &= P_X - P_X P_X \\ &= P_X - P_X \\ &= 0. \end{aligned}$$

This property implies that $M_X X = 0$. Indeed,

$$\begin{aligned} M_X X &= (I_n - P_X) X \\ &= X - P_X X \\ &= X - X \\ &= 0. \end{aligned}$$

In the above discussion, none of the regression assumptions have been used. Given data y and X, one can always perform least squares, regardless of the data-generating process. However, a model is needed to discuss the statistical properties of an estimator (such as unbiasedness).

Properties of $\widehat{\sigma}^2$

The following estimator of σ^2 was introduced in Lecture 2:

$$\begin{aligned} \widehat{\sigma}^2 &= n^{-1} \sum_{i=1}^{n} \left(Y_i - X_i^\top \widehat{\beta} \right)^2 \\ &= n^{-1} \widehat{U}^\top \widehat{U}. \end{aligned}$$

Under Assumptions (A1)–(A4), $\widehat{\sigma}^2$ is biased. First, write

$$\begin{aligned} \widehat{U} &= M_X Y \\ &= M_X (X\beta + U) \\ &= M_X U. \end{aligned}$$

The last equality follows because $M_X X = 0$. Next,

$$\begin{aligned} n\widehat{\sigma}^2 &= \widehat{U}^\top \widehat{U} \\ &= U^\top M_X M_X U \\ &= U^\top M_X U. \end{aligned}$$

Now, since $U^\top M_X U$ is a scalar,

$$U^\top M_X U = \operatorname{tr}\left(U^\top M_X U\right),$$

where $\operatorname{tr}(A)$ denotes the trace of a matrix A.

$$\begin{aligned} \mathrm{E}\left[U^\top M_X U \mid X\right] &= \mathrm{E}\left[\operatorname{tr}\left(U^\top M_X U\right) \mid X\right] \\ &= \mathrm{E}\left[\operatorname{tr}\left(M_X U U^\top\right) \mid X\right] \quad (\text{because } \operatorname{tr}(ABC) = \operatorname{tr}(BCA)) \\ &= \operatorname{tr}\left(M_X \, \mathrm{E}\left[U U^\top \mid X\right]\right) \quad (\text{because } \operatorname{tr} \text{ and expectation are linear operators}) \\ &= \sigma^2 \operatorname{tr}(M_X). \end{aligned}$$

The last equality follows because, by Assumption (A3), $\mathrm{E}\left[UU^\top \mid X\right] = \sigma^2 I_n$. Next,

$$\begin{aligned} \operatorname{tr}(M_X) &= \operatorname{tr}\left(I_n - X\left(X^\top X\right)^{-1} X^\top\right) \\ &= \operatorname{tr}(I_n) - \operatorname{tr}\left(X\left(X^\top X\right)^{-1} X^\top\right) \\ &= \operatorname{tr}(I_n) - \operatorname{tr}\left(\left(X^\top X\right)^{-1} X^\top X\right) \\ &= \operatorname{tr}(I_n) - \operatorname{tr}(I_k) \\ &= n - k. \end{aligned}$$

Thus,

$$\mathrm{E}[\widehat{\sigma}^2] = \frac{n-k}{n}\sigma^2. \tag{2}$$

The estimator $\widehat{\sigma}^2$ is biased, but a simple modification yields an unbiased estimator. Define

$$\begin{aligned} s^2 &= \widehat{\sigma}^2 \frac{n}{n-k} \\ &= (n-k)^{-1} \sum_{i=1}^{n} \left(Y_i - X_i^\top \widehat{\beta} \right)^2. \end{aligned}$$

It follows from (2) that

$$\mathrm{E}[s^2] = \sigma^2.$$

Partitioned regression

We can partition the matrix of regressors X as follows:

$$X = \begin{pmatrix} X_1 & X_2 \end{pmatrix},$$

and write the model as

$$Y = X_1\beta_1 + X_2\beta_2 + U,$$

where X_1 is an $n \times k_1$ matrix, X_2 is $n \times k_2$, $k_1 + k_2 = k$, and

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

where β_1 and β_2 are k_1- and k_2-vectors, respectively. Such a decomposition allows one to focus on a group of variables and their corresponding parameters, say X_1 and β_1. If

$$\widehat{\beta} = \begin{pmatrix} \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{pmatrix},$$

then one can write the following version of the normal equations:

$$\left(X^\top X \right) \widehat{\beta} = X^\top Y$$

as

$$\begin{pmatrix} X_1^\top X_1 & X_1^\top X_2 \\ X_2^\top X_1 & X_2^\top X_2 \end{pmatrix} \begin{pmatrix} \widehat{\beta}_1 \\ \widehat{\beta}_2 \end{pmatrix} = \begin{pmatrix} X_1^\top Y \\ X_2^\top Y \end{pmatrix}.$$

The expressions for $\widehat{\beta}_1$ and $\widehat{\beta}_2$ can be obtained by inverting the partitioned matrix on the left-hand side.

Alternatively, define M_2 to be the projection matrix onto the space orthogonal to $\mathcal{S}(X_2)$:

$$M_2 = I_n - X_2 \left(X_2^\top X_2 \right)^{-1} X_2^\top.$$

Then,

$$\widehat{\beta}_1 = \left(X_1^\top M_2 X_1 \right)^{-1} X_1^\top M_2 Y. \tag{3}$$

To show this, first write

$$Y = X_1\widehat{\beta}_1 + X_2\widehat{\beta}_2 + \widehat{U}. \tag{4}$$

By construction,

$$\begin{aligned}
M_2\widehat{U} &= \widehat{U} \text{ } (\widehat{U} \text{ is orthogonal to } X_2), \\
M_2 X_2 &= 0, \\
X_1^\top \widehat{U} &= 0, \\
X_2^\top \widehat{U} &= 0.
\end{aligned}$$

Substitute equation (4) into the right-hand side of equation (3):

$$\begin{aligned}
&\left(X_1^\top M_2 X_1\right)^{-1} X_1^\top M_2 Y \\
&= \left(X_1^\top M_2 X_1\right)^{-1} X_1^\top M_2 \left(X_1\widehat{\beta}_1 + X_2\widehat{\beta}_2 + \widehat{U}\right) \\
&= \left(X_1^\top M_2 X_1\right)^{-1} X_1^\top M_2 X_1 \widehat{\beta}_1 \\
&+ \left(X_1^\top M_2 X_1\right)^{-1} X_1^\top \widehat{U} \qquad \left(M_2 X_2 = 0 \text{ and } M_2\widehat{U} = \widehat{U}\right) \\
&= \widehat{\beta}_1.
\end{aligned}$$

Since M_2 is symmetric and idempotent, one can write

$$\begin{aligned}
\widehat{\beta}_1 &= \left(\left(M_2 X_1\right)^\top \left(M_2 X_1\right)\right)^{-1} \left(M_2 X_1\right)^\top \left(M_2 Y\right) \\
&= \left(\widetilde{X}_1^\top \widetilde{X}_1\right)^{-1} \widetilde{X}_1^\top \widetilde{Y},
\end{aligned}$$

where

$$\begin{aligned}
\widetilde{X}_1 &= M_2 X_1 \\
&= X_1 - X_2 \left(X_2^\top X_2\right)^{-1} X_2^\top X_1 \text{ (residuals from the regression of } X_1 \text{ on } X_2), \\
\widetilde{Y} &= M_2 Y \\
&= Y - X_2 \left(X_2^\top X_2\right)^{-1} X_2^\top Y \text{ (residuals from the regression of } Y \text{ on } X_2).
\end{aligned}$$

Thus, to obtain the coefficients on the first k_1 regressors, instead of running the full regression with k_1+k_2 regressors, one can regress Y on X_2 to obtain the residuals $\widetilde{Y}$, regress X_1 on X_2 to obtain the residuals $\widetilde{X}_1$, and then regress $\widetilde{Y}$ on $\widetilde{X}_1$ to obtain $\widehat{\beta}_1$. In other words, $\widehat{\beta}_1$ shows the effect of X_1 *after controlling* for X_2.

Analogously, for $\widehat{\beta}_2$:

$$\begin{aligned}
\widehat{\beta}_2 &= \left(X_2^\top M_1 X_2\right)^{-1} X_2^\top M_1 Y, \text{ where} \\
M_1 &= I_n - X_1 \left(X_1^\top X_1\right)^{-1} X_1^\top.
\end{aligned}$$

For example, consider a simple regression

$$Y_i = \beta_1 + \beta_2 X_i + U_i,$$

for $i = 1, \ldots, n$.

Define an n-vector of ones:

$$\ell = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix}.$$

In this case, the matrix of regressors is given by

$$\begin{pmatrix} 1 & X_1 \\ 1 & X_2 \\ \vdots & \vdots \\ 1 & X_n \end{pmatrix} = \begin{pmatrix} \ell & X \end{pmatrix}.$$

Consider

$$M_1 = I_n - \ell \left(\ell^\top \ell\right)^{-1} \ell^\top,$$

and

$$\widehat{\beta}_2 = \frac{X^\top M_1 Y}{X^\top M_1 X}.$$

Now, $\ell^\top \ell = n$. Therefore,

$$\begin{aligned} M_1 &= I_n - \frac{1}{n}\ell\ell^\top, \text{ and} \\ M_1 X &= X - \ell\frac{\ell^\top X}{n} \\ &= X - \overline{X}\ell \\ &= \begin{pmatrix} X_1 - \overline{X} \\ X_2 - \overline{X} \\ \vdots \\ X_n - \overline{X} \end{pmatrix}, \end{aligned}$$

where

$$\begin{aligned} \overline{X} &= \frac{\ell^\top X}{n} \\ &= n^{-1}\sum_{i=1}^{n} X_i. \end{aligned}$$

Thus, the matrix M_1 transforms the vector X into the vector of deviations from the average. We can write

$$\begin{aligned} \widehat{\beta}_2 &= \frac{\sum_{i=1}^n \left(X_i - \overline{X}\right) Y_i}{\sum_{i=1}^n \left(X_i - \overline{X}\right)^2} \\ &= \frac{\sum_{i=1}^n \left(X_i - \overline{X}\right)\left(Y_i - \overline{Y}\right)}{\sum_{i=1}^n \left(X_i - \overline{X}\right)^2}. \end{aligned}$$

Goodness of fit

Write

$$\begin{aligned} Y &= P_X Y + M_X Y \\ &= \widehat{Y} + \widehat{U}, \end{aligned}$$

where, by construction,

$$\begin{aligned} \widehat{Y}^\top \widehat{U} &= \left(P_X Y\right)^\top \left(M_X Y\right) \\ &= Y^\top P_X M_X Y \\ &= 0. \end{aligned}$$

Suppose that the model contains an intercept, i.e., the first column of X is the vector of ones ℓ. The *total variation* in Y is

$$\begin{aligned}\sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2 &= Y^\top M_1 Y \\ &= \left(\widehat{Y}+\widehat{U}\right)^\top M_1\left(\widehat{Y}+\widehat{U}\right) \\ &= \widehat{Y}^\top M_1\widehat{Y}+\widehat{U}^\top M_1\widehat{U}+2\widehat{Y}^\top M_1\widehat{U}.\end{aligned}$$

Since the model contains an intercept,

$$\begin{aligned}\ell^\top\widehat{U} &= 0, \text{ and} \\ M_1\widehat{U} &= \widehat{U}.\end{aligned}$$

However, $\widehat{Y}^\top\widehat{U}=0$, and, therefore,

$$\begin{aligned}Y^\top M_1 Y &= \widehat{Y}^\top M_1\widehat{Y}+\widehat{U}^\top\widehat{U}, \text{ or} \\ \sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2 &= \sum_{i=1}^{n}\left(\widehat{Y}_i-\overline{\widehat{Y}}\right)^2+\sum_{i=1}^{n}\widehat{U}_i^2.\end{aligned}$$

Note that

$$\begin{aligned}\overline{Y} &= \frac{\ell^\top Y}{n} \\ &= \frac{\ell^\top\widehat{Y}}{n}+\frac{\ell^\top\widehat{U}}{n} \\ &= \frac{\ell^\top\widehat{Y}}{n} \\ &= \overline{\widehat{Y}}.\end{aligned}$$

Hence, the averages of Y and its predicted values $\widehat{Y}$ are equal, and we can write:

$$\begin{aligned}\sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2 &= \sum_{i=1}^{n}\left(\widehat{Y}_i-\overline{Y}\right)^2+\sum_{i=1}^{n}\widehat{U}_i^2, \text{ or} \\ TSS &= ESS+RSS,\end{aligned} \tag{5}$$

where

$$\begin{aligned}TSS &= \sum_{i=1}^{n}\left(Y_i-\overline{Y}\right)^2 \text{ (total sum of squares)}, \\ ESS &= \sum_{i=1}^{n}\left(\widehat{Y}_i-\overline{Y}\right)^2 \text{ (explained sum of squares)}, \\ RSS &= \sum_{i=1}^{n}\widehat{U}_i^2 \text{ (residual sum of squares)}.\end{aligned}$$

The ratio of ESS to TSS is called the *coefficient of determination*, or R^2:

$$
\begin{aligned}
R^2 &= \frac{\sum_{i=1}^{n}\left(\widehat{Y}_i - \overline{Y}\right)^2}{\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2} \\
&= 1 - \frac{\sum_{i=1}^{n}\widehat{U}_i^2}{\sum_{i=1}^{n}\left(Y_i - \overline{Y}\right)^2} \\
&= 1 - \frac{\widehat{U}^\top \widehat{U}}{Y^\top M_1 Y}.
\end{aligned}
$$

Properties of R^2:

- R^2 is bounded between 0 and 1, as implied by decomposition (5). This property does not hold if the model lacks an intercept; one should not use the above definition of R^2 in that case. If $R^2 = 1$, then $\widehat{U}^\top \widehat{U} = 0$, which can happen only if $Y \in \mathcal{S}(X)$, i.e., Y is *exactly* a linear combination of the columns of X.

- R^2 increases when additional regressors are added.

  Proof. Consider a partitioned matrix $X = \begin{pmatrix} Z & W \end{pmatrix}$. We study the effect of adding W on R^2. Let

  $$
  \begin{aligned}
  P_X &= X\left(X^\top X\right)^{-1} X^\top \text{ (projection matrix for the full regression)}, \\
  P_Z &= Z\left(Z^\top Z\right)^{-1} Z^\top \text{ (projection matrix for the regression without } W\text{)}.
  \end{aligned}
  $$

  Define also

  $$
  \begin{aligned}
  M_X &= I_n - P_X, \\
  M_Z &= I_n - P_Z.
  \end{aligned}
  $$

  Since Z is part of X,

  $$P_X Z = Z,$$

  and

  $$
  \begin{aligned}
  P_X P_Z &= P_X Z\left(Z^\top Z\right)^{-1} Z^\top \\
  &= Z\left(Z^\top Z\right)^{-1} Z^\top \\
  &= P_Z.
  \end{aligned}
  $$

  Consequently,

  $$
  \begin{aligned}
  M_X M_Z &= \left(I_n - P_X\right)\left(I_n - P_Z\right) \\
  &= I_n - P_X - P_Z + P_X P_Z \\
  &= I_n - P_X - P_Z + P_Z \\
  &= M_X.
  \end{aligned}
  $$

  Assume that Z contains a column of ones, so both short and long regressions have intercepts. Define

  $$
  \begin{aligned}
  \widehat{U}_X &= M_X Y, \\
  \widehat{U}_Z &= M_Z Y.
  \end{aligned}
  $$

  Write:

  $$
  \begin{aligned}
  0 &\le \left(\widehat{U}_X - \widehat{U}_Z\right)^\top \left(\widehat{U}_X - \widehat{U}_Z\right) \\
  &= \widehat{U}_X^\top \widehat{U}_X + \widehat{U}_Z^\top \widehat{U}_Z - 2\widehat{U}_X^\top \widehat{U}_Z.
  \end{aligned}
  $$

Next,

$$\begin{aligned}\widehat{U}_X^\top \widehat{U}_Z &= Y^\top M_X M_Z Y \\ &= Y^\top M_X Y \\ &= \widehat{U}_X^\top \widehat{U}_X.\end{aligned}$$

Hence,

$$\widehat{U}_Z^\top \widehat{U}_Z \geq \widehat{U}_X^\top \widehat{U}_X.$$

- R^2 shows how much of the *sample* variation in Y was explained by X. However, our objective is to estimate *population* relationships, not to explain *sample* variation. A high R^2 does not necessarily indicate a good regression model, and a low R^2 is not evidence against one.
- One can always find an X that makes $R^2 = 1$: take any n linearly independent vectors. Since such a set spans $\mathbb{R}^n$, any $y \in \mathbb{R}^n$ can be written as an exact linear combination of the columns of X.

Since R^2 increases with the inclusion of additional regressors, researchers often report instead the *adjusted coefficient of determination* $\overline{R}^2$:

$$\begin{aligned}\overline{R}^2 &= 1 - \frac{n-1}{n-k}\left(1 - R^2\right) \\ &= 1 - \frac{\widehat{U}^\top \widehat{U} / (n-k)}{Y^\top M_1 Y / (n-1)}.\end{aligned}$$

The adjusted coefficient of determination penalizes the fit when the number of regressors k is large relative to the number of observations n, and $\overline{R}^2$ may decrease as k increases. However, there is no strong argument for using this adjustment.